Pandrosion Prime Atlases and the Riemann Critical Line

Euler factors as infinite geometric charts

Ivan Besevic
Independent Mathematical Research

May 2026

Abstract

This paper proposes a new way to connect the geometric Pandrosion method of Paper 000 with the Riemann zeta function. The connection is not the finite identity $\sum_k 1/P'(\lambda_k) = 0$, which is classical Lagrange interpolation. The connection is instead the logarithmic chart geometry behind the Euler product. Paper 000 shows that the monomial map for $z^p = x$ is governed by the finite geometric sum

$$S_p(s) = 1 + s + \cdots + s^{p-1},$$

and that the correct geometry first chooses a Thales homothety and, when needed, a reciprocal Riemann chart so that a normalized target lies close to 1. For the zeta function, every Euler factor is an infinite geometric Pandrosion cell:

$$\frac{1}{1-\ell^{-s}} = 1 + \ell^{-s} + \ell^{-2s} + \cdots .$$

We therefore define a prime-by-prime Pandrosion atlas with direct chart $q_\ell(s) = \ell^{-s}$ and reciprocal chart $q_\ell^\infty(s) = \ell^{-(1-s)}$. The involution $s \mapsto 1-s$, which is the symmetry axis of the completed zeta functional equation, becomes the global version of the Riemann-chart inversion from Paper 000.

The main rigorous observation is finite and geometric. For any finite set of primes and any positive weights, the logarithmic direct and reciprocal chart costs

$$\mathcal{E}_+(\sigma) = \sum_\ell w_\ell(\sigma \log \ell)^2, \qquad \mathcal{E}_\infty(\sigma) = \sum_\ell w_\ell((1-\sigma)\log \ell)^2$$

are balanced at exactly $\sigma = 1/2$. Thus the critical line is the unique Thales–Riemann equilibrium line of the prime Pandrosion atlas. This is not a proof of the Riemann hypothesis. It is a new geometric formulation: the critical line is where the direct Euler charts and reciprocal Euler charts have equal logarithmic distortion, and the nontrivial zeros may be viewed as phase destruction events on that balanced atlas.

One-sentence summary. Euler factors are infinite Pandrosion geometric sums, and the critical line $\Re(s) = 1/2$ is the chart-equilibrium locus between the direct prime chart ℓ^{-s} and the reciprocal prime chart $\ell^{-(1-s)}$.

Status. The atlas construction and critical-line balance theorem are elementary and proved below. The proposed interpretation of zeta zeros as balanced-atlas phase destruction is a research program, not a proof of the Riemann hypothesis.

Contents

1 Motivation

The legacy Riemann note used Pandrosion language for random matrices, zeta zeros, and polynomial spectral identities. Some of its formulas are useful as notation, but many of the algebraic identities are classical. For example, if $P(z) = \prod_k (z - \lambda_k)$ has simple roots, then

$$P'(\lambda_k) = \prod_{j \neq k} (\lambda_k - \lambda_j), \qquad \sum_k \frac{1}{P'(\lambda_k)} = 0$$

for degree at least 2. This is not a new route to the zeta function.

The more distinctive Pandrosion structure in Paper 000 is different. It is the combination of:

1. finite geometric sums S_p;
2. logarithmic normalization by homothety;
3. reciprocal Riemann-chart inversion;
4. choosing the chart that keeps $|\log y|$ small.

The Riemann zeta function is naturally built from logarithmic scales:

$$\zeta(s) = \prod_{\ell \text{ prime}} \frac{1}{1 - \ell^{-s}}.$$

This suggests using the Euler product itself as the bridge from Pandrosion geometry to zeta.

2 From monomial Pandrosion to Euler cells

In Paper 000 the basic monomial denominator is

$$S_p(s) = 1 + s + \cdots + s^{p-1}.$$

For a prime ℓ and a complex variable s, put

$$q_\ell(s) = \ell^{-s}.$$

The finite Euler-Pandrosion cell of depth N is

$$E_{\ell,N}(s) = S_N(q_\ell(s)) = 1 + \ell^{-s} + \ell^{-2s} + \cdots + \ell^{-(N-1)s}.$$

For $\Re(s) > 0$, $E_{\ell,N}(s)$ converges as $N \to \infty$ to the Euler factor:

$$E_{\ell,\infty}(s) = \frac{1}{1 - \ell^{-s}}.$$

Definition 2.1 (Euler-Pandrosion product)**.** *For a finite prime set $\mathcal{P}$ and depth N, define*

$$\mathcal{Z}_{\mathcal{P},N}(s) = \prod_{\ell \in \mathcal{P}} E_{\ell,N}(s).$$

When $N = \infty$ and $\mathcal{P}$ increases over all primes, this tends to $\zeta(s)$ in the half-plane $\Re(s) > 1$.

Thus zeta is not merely analogous to Pandrosion. In the Euler half-plane it is a limiting product of infinite Pandrosion geometric cells.

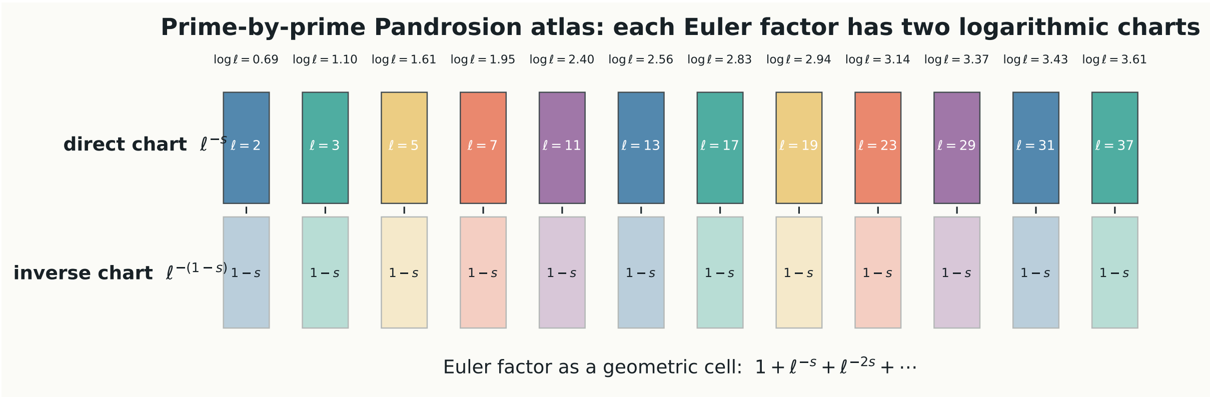

Figure 1: Prime-by-prime Pandrosion atlas. Each prime contributes a direct Euler cell $1 + \ell^{-s} + \ell^{-2s} + \cdots$ and a reciprocal cell obtained by the involution $s \mapsto 1 - s$.

3 The reciprocal chart and the functional-equation axis

The reciprocal Riemann chart in Paper 000 moves a small or badly scaled target into a normalized chart before iteration. For Euler factors the analogous operation is the involution

$$s \longmapsto 1 - s.$$

It sends the direct prime coordinate $q_\ell(s) = \ell^{-s}$ to the reciprocal prime coordinate

$$q_\ell^\infty(s) = \ell^{-(1-s)}.$$

This is not yet the full zeta functional equation; the completed zeta function

$$\xi(s) = \frac{1}{2} s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$$

also contains the gamma and conductor terms. The point is geometric: the same involution that underlies the completed functional equation is the global prime-atlas version of the Riemann-chart inversion in the monomial paper.

Definition 3.1 (Prime-atlas chart costs)**.** *Let $\mathcal{P}$ be a finite set of primes and $w_\ell > 0$. For $\sigma = \Re(s)$, define*

$$\mathcal{E}_+(\sigma) = \sum_{\ell \in \mathcal{P}} w_\ell (\sigma \log \ell)^2, \qquad \mathcal{E}_\infty(\sigma) = \sum_{\ell \in \mathcal{P}} w_\ell ((1-\sigma)\log \ell)^2.$$

The selected atlas cost is

$$\mathcal{E}_{\mathrm{sel}}(\sigma) = \min(\mathcal{E}_+(\sigma), \mathcal{E}_\infty(\sigma)).$$

The squared logarithmic costs are the direct analogues of the $|\log y|$-based chart rule in Paper 000. The direct chart measures how far ℓ^{-s} is from unit scale; the inverse chart measures the same quantity after $s \mapsto 1 - s$.

Theorem 3.2 (Critical line as Pandrosion chart equilibrium)**.** *For every finite nonempty prime set $\mathcal{P}$ and every positive weight family (w_ℓ), the equality*

$$\mathcal{E}_+(\sigma) = \mathcal{E}_\infty(\sigma)$$

holds if and only if $\sigma = 1/2$. Equivalently, the line $\Re(s) = 1/2$ is the unique direct/reciprocal chart-equilibrium locus of the prime Pandrosion atlas.

Proof. Subtract the two costs:

$$\mathcal{E}_+(\sigma) - \mathcal{E}_\infty(\sigma) = \sum_{\ell\in\mathcal{P}} w_\ell \left(\sigma^2 - (1-\sigma)^2\right) (\log \ell)^2.$$

Since

$$\sigma^2 - (1-\sigma)^2 = 2\sigma - 1,$$

we get

$$\mathcal{E}_+(\sigma) - \mathcal{E}_\infty(\sigma) = (2\sigma - 1) \sum_{\ell\in\mathcal{P}} w_\ell (\log \ell)^2.$$

The sum is strictly positive. Hence equality is equivalent to $2\sigma - 1 = 0$, i.e. $\sigma = 1/2$. $\square$

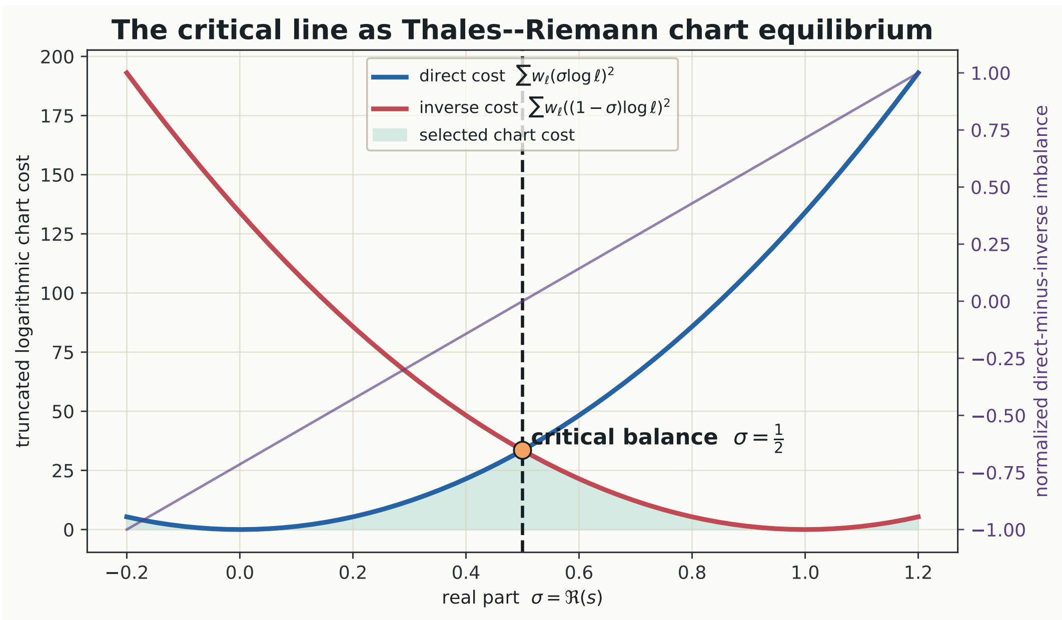

Figure 2: The critical line as chart equilibrium. Direct and reciprocal logarithmic distortions are equal exactly at $\sigma = 1/2$, independently of the positive weights assigned to the finite prime atlas.

4 Phase, cancellation, and zeros

The balance theorem explains why the critical line is geometrically special, but it does not locate zeros. Zeros require phase cancellation. Write

$$q_\ell(\sigma + it) = \ell^{-\sigma} e^{-it\log\ell}.$$

The modulus is controlled by the chart coordinate σ; the phase rotates with frequency $\log \ell$. The primes therefore generate an incommensurate family of rotating geometric cells.

Definition 4.1 (Balanced Euler-Pandrosion phase field)**.** *For a finite prime set and finite depth, define*

$$\Phi_{\mathcal{P},N}(s) = \arg \prod_{\ell \in \mathcal{P}} E_{\ell,N}(s),$$

away from zeros of the truncated product. On the critical line this becomes

$$\Phi_{\mathcal{P},N}(1/2 + it).$$

In this language, a zero of a completed limiting object is not a mysterious point attached to a finite interpolation identity. It is a destructive phase event in an infinite balanced geometric atlas.

Conjecture 4.2 (Pandrosion prime-atlas cancellation principle)**.** *After the gamma/conductor normalization required by the completed zeta function, nontrivial zeta zeros are limiting phase-destruction events of the balanced prime Pandrosion atlas. The line $\Re(s) = 1/2$ is selected not by a finite polynomial vanishing identity, but by the equality of direct and reciprocal logarithmic chart distortions prime by prime.*

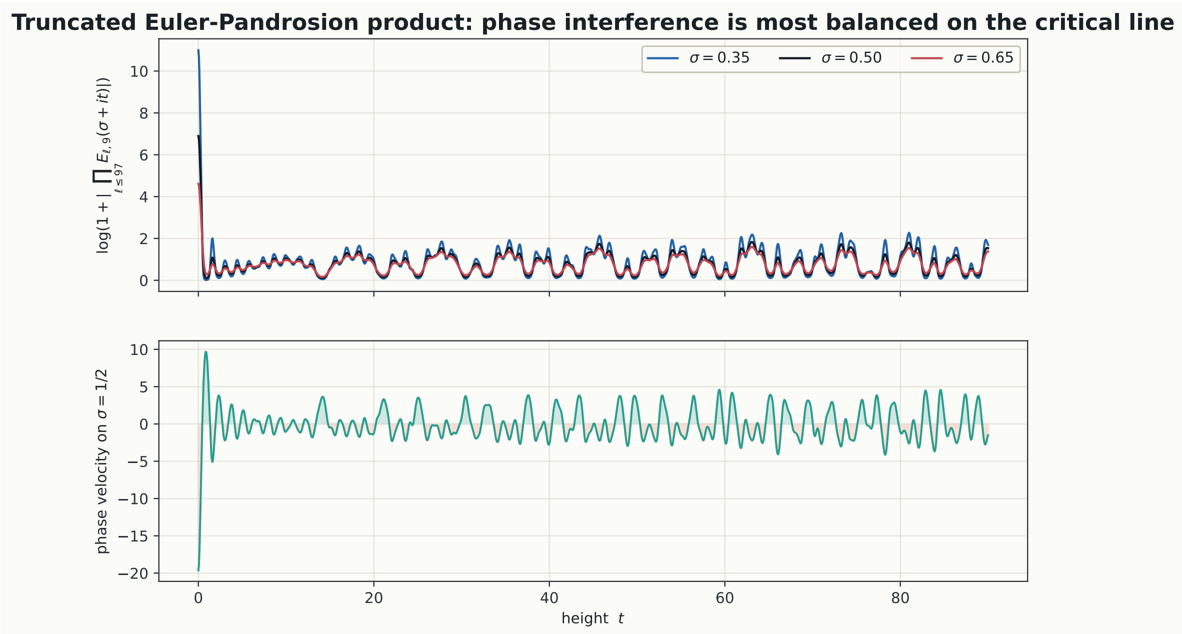

Figure 3: A finite numerical illustration, not evidence for the Riemann hypothesis. The product $\prod_{\ell \le 97} E_{\ell,9}(\sigma + it)$ shows phase interference from the prime logarithms. The critical-line trace is the balanced direct/reciprocal chart slice.

5 Relation to the completed zeta function

The Euler product alone converges only in $\Re(s) > 1$. Any serious relation to the critical strip must include the completed function

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \xi(s) = \xi(1-s).$$

In the present framework the gamma/conductor factor should be treated as the archimedean chart, while the Euler factors are the non-archimedean prime charts. Thus the full object is not a single prime product; it is an adelic-style Pandrosion atlas:

$$\text{archimedean chart} \quad + \quad \text{prime geometric charts}.$$

The finite theorem above explains the prime-side equilibrium. The completed functional equation says that the archimedean chart respects the same involution.

6 What is new and what is not

The following distinctions are important.

1. The identity $\sum_k 1/P'(\lambda_k) = 0$ is classical and should not be advertised as a new formula.
2. Euler products and the functional equation of ζ are classical.
3. The new proposal is the geometric reading: Euler factors are infinite Pandrosion sums, $s \mapsto 1 - s$ is the global Riemann-chart inversion, and $\Re(s) = 1/2$ is the unique logarithmic chart-equilibrium line of the prime atlas.
4. The zero interpretation is conjectural. It reframes the problem in terms of balanced-atlas phase cancellation; it does not prove that all zeros lie on the critical line.

7 Research program

This viewpoint suggests concrete next steps.

1. Add the archimedean gamma chart explicitly and define a completed Pandrosion atlas whose involution is exactly $s \mapsto 1 - s$.
2. Replace the elementary quadratic chart cost by a natural energy derived from $\log |\xi(s)|$ or from the argument principle.
3. Study whether off-critical points have unavoidable direct/reciprocal chart imbalance after completion.
4. Compare the phase velocity of truncated Euler-Pandrosion products with known zero-counting formulas and the Riemann–Siegel theta term.
5. Revisit CUE/GUE statistics only after the prime-atlas construction is in place, so random matrices model balanced-atlas phase statistics rather than replace the Pandrosion geometry.

8 Conclusion

The cleanest link between Paper 000 and the Riemann zeta function is not the finite spectral identity of the legacy Riemann note. It is the fact that every Euler factor is an infinite geometric Pandrosion cell and that the zeta functional equation uses the same direct/reciprocal involution as the Riemann-chart logic of monomial Pandrosion. In this prime-atlas geometry, the critical line is not inserted by hand: it is the unique locus where direct and reciprocal logarithmic chart costs are equal. This gives a genuinely geometric Pandrosion formulation of the critical line, and a precise research program for turning phase-cancellation heuristics into testable mathematics.

References

[1] I. Besevic, *A Monomial Pandrosion Scheme: Contraction, Chart Scaling, and Steffensen Acceleration for $z^p = x$*, preprint, May 2026.

[2] I. Besevic, *Chart-Scaled Cubic Pandrosion by Finite-Slope Halley Acceleration*, preprint, May 2026.

[3] I. Besevic, *Pandrosion Atlases and Dynamic Geometric Charts*, preprint, May 2026.

[4] B. Riemann, "Ueber die Anzahl der Primzahlen unter einer gegebenen Grosse," *Monatsberichte der Berliner Akademie*, 1859.

[5] E. C. Titchmarsh, *The Theory of the Riemann Zeta-Function*, 2nd ed., revised by D. R. Heath-Brown, Oxford University Press, 1986.

[6] H. M. Edwards, *Riemann's Zeta Function*, Dover, 2001.

[7] A. Ivic, *The Riemann Zeta-Function: Theory and Applications*, Dover, 2003.

[8] H. L. Montgomery, "The pair correlation of zeros of the zeta function," *Proc. Sympos. Pure Math.* 24, 181–193, 1973.

[9] J. P. Keating and N. C. Snaith, "Random matrix theory and $\zeta(1/2 + it)$," *Communications in Mathematical Physics* 214, 57–89, 2000.